

Code-Layout, Readability and Reusability

Date	3 July 2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used Throughout the code	Yes	Proper indentation and formatting maintained throughout the Python, Flask, HTML, and CSS files.
2	Proper File Structure	Files and folders are logically organized	Yes	Project organized into templates, static, models, dataset, and application files for better maintainability.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables such as loan_amount, credit_history, annual_income, and prediction_result clearly indicate their purpose.
4	Function / Method Names	Functions are descriptively named	Yes	Functions like predict(), load_model(), preprocess_data(), and train_model() are descriptive and easy to understand.
5	Code Comments	Inline and block comments explain logic	Yes	Important sections include comments explaining preprocessing, model training, prediction logic, and routing.
6	Modular Design	Code is split into reusable functions/modules	Yes	Data preprocessing, model training, prediction, and web interface are implemented as separate modules/functions.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate code removed by reusing functions and maintaining a clean project structure.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation and exception handling ensure reliable predictions.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technolog	yWhere Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing Module	Python, Pandas	Training and Prediction	High
2	Model Training Module	Scikit-learn, XGBoost	Model Development	High
3	Prediction Module	Python, Flask	Loan Approval Prediction	High
4	Flask Routes	Flask	User Request Handling	High
5	HTML Templates	HTML, CSS, Bootstrap	Web Interface	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-structured project with organized folders and modular design.
Readability	5	Clear naming conventions and readable code throughout the project.
Reusability	5	Reusable functions and modular components improve maintainability.
Documentation / Comments	4	Important functions are documented with appropriate comments.
Overall Score	4.75/5	Clean, maintainable, and reusable code following good coding practices.